P2 Group 02 Database Design Document

Business Problems Addressed

Service-based businesses such as salons, medspas, and pet groomers often rely on manual or in-person scheduling, such as **phone calls, walk-ins, or paper calendars.** This creates several problems:

- **Limited booking access.** Customers can only book during business hours and must call or visit in person. This system lets them book, modify, and cancel appointments digitally at any time.
- **Scheduling conflicts and double-booking.** Without a central record of staff availability, two customers can be booked with the same staff member at the same time. The system generates bookable time slots from each staff member's actual availability and reserves a slot once it is booked.
- **Matching customers with qualified staff.** Not every employee can perform every service. The system records which staff members can perform which services, along with their proficiency level, so customers are only booked with someone qualified.
- **Lost revenue from late cancellations and no-shows.** Unfilled slots cost businesses money. The system lets each business define its own cancellation policy (notice period, fee, and refund rule), records every cancellation, and links appointments to payments so fees can be applied consistently.
- **Payment handling.** Customers need a convenient way to pay deposits or service charges, and repeat customers benefit from stored payment methods. The system tracks service pricing, estimated charges, and payment transactions.
- **Poor communication.** Customers forget appointments or are unsure whether a booking went through. The system sends notifications such as confirmations and reminders.
- **Staff time spent on logistics.** By moving scheduling online, front-desk staff can spend less time managing calendars and more time with clients on site.

The design is scoped narrowly around the booking and transaction flow. It is built so that future features, such as sales reporting and customer reviews, can be added without restructuring the core entities.

1. Entities and Their Relationships

The database contains 14 entities and one relationship with its own attribute (ProficiencyLevel).

Entity	Purpose	Key Attributes
BUSINESS	The company that owns the brand and offers services	BusinessID, Name, Description, Website, Social Media
LOCATION	Where the company is located	LocationID, Address, OperatingHours, ContactPhone, Website, Social Media {Facebook, Instagram, etc}
STAFF	An employee who performs services	StaffID, Name, Role
STAFF_AVAILABILITY	A staff member's recurring working pattern	Date, StartTime,Endtime
TIME_SLOT	A specific bookable block of time	TimeSlotID, StartTime, EndTime
SERVICE_CATEGORY	A group of related services (e.g., Nails, Pet Grooming)	ServiceCategoryID, Name, Description
SERVICE	A specific service offered (e.g., Manicure)	ServiceID, Name, Description, EstimatedDuration, Price
CUSTOMER	A person who books appointments	CustomerID, Name, PhoneNumber, Email, RegistrationDate, CustomerPreferredContact
APPOINTMENT	A customer's booking for a service	AppointmentID, ScheduledDateTime, Status {Confirmed, Waitlisted,Cancelled,Completed}, EstimatedCharge

CANCELLATION_POLICY	A business's rules for cancellations	NoticePeriod, FeeAmount, RefundRule
CANCELLATION	A record of a cancelled appointment	CancellationID, CancelledAt, CancelledBy, Reason, FeeCharged, RefundAmount
PAYMENT_METHOD	A stored payment method (e.g., card)	PaymentMethodID, CardType {Visa, MasterCard, American Express}, LastFourDigits, ExpirationDate
PAYMENT	A payment transaction for an appointment	PaymentID, Amount, Status {Pending, Charged,Refunded,Waived, Failed}, ChargedAt, Type
NOTIFICATION	A message sent to a customer	NotificationID, Type {Confirmation, Reminder, Cancellation Alert}, SentAt, DeliveryMethod

Relationships

Relationships	Entities	Type	Description
operates	BUSINESS → LOCATION	One-to-one	A business operates one location; the location belongs to one business.
defines	BUSINESS → SERVICE_CATEGORY	One-to-many	A business defines the categories of services it offers.
sets	BUSINESS → CANCELLATION_POLICY	One-to-one	A business sets the cancellation policy that

			applies to its appointments.
employs	LOCATION → STAFF	One-to-many	A location employs multiple staff members.
groups	SERVICE_CATEGORY → SERVICE	One-to-many	A category groups one or more services.
has	STAFF → STAFF_AVAILABILITY	One-to-many	A staff member has multiple availability entries (e.g., different days).
generates	STAFF_AVAILABILITY → TIME_SLOT	One-to-many	Each availability entry generates the time slots customers can book.
performs	STAFF ↔ SERVICE	Many-to-many, then converted to one-to-many with StaffService as associative entity. StaffService is one-to-many with Service	Staff can perform many services, and each service can be performed by many staff. The relationship stores a proficiency level.
books	CUSTOMER → APPOINTMENT	One-to-many	A customer can book many appointments over time.

booked as	SERVICE → APPOINTMENT	One-to-many. Connected via associative entity Appointment service.	Each appointment is for a specific service
assigned to	STAFF → APPOINTMENT	One-to-many	Each appointment is assigned to one staff member.
reserved by	TIME_SLOT → APPOINTMENT	One-to-many	A time slot is reserved by an appointment.
governs	CANCELLATION_POLICY → APPOINTMENT	One-to-many	A policy governs the appointments booked under it.
secured by	APPOINTMENT → PAYMENT	One-to-one	An appointment is secured by one or more payments (e.g., deposit, final charge, fee).
results in	APPOINTMENT → CANCELLATION	One-to-zero-or-one	An appointment may result in a cancellation, but most do not.
triggers	APPOINTMENT → NOTIFICATION	One-to-many	An appointment triggers notifications such as confirmations and reminders.
stores	CUSTOMER → PAYMENT_METHOD	One-to-many	A customer can store

			several payment methods.
charges	PAYMENT_METHOD → PAYMENT	One-to-many	A payment method is charged for payments; a method can be used many times.
receives	CUSTOMER → NOTIFICATION	One-to-many	A customer receives notifications.

Associative entities

StaffService

StaffService is the associative entity that resolves the many-to-many relationship between Staff and Service — which staff members can perform which services. It carries proficiencyLevel, capturing how skilled a given staff member is at a given service.

A link from StaffService to CancellationPolicy was considered briefly, to let cancellation fees vary by staff proficiency (e.g., a smaller fee for a beginner, a higher fee for a highly proficient staff member). That link has been removed now that the business uses a single cancellation policy overall — StaffService is back to representing only Staff, Service, and proficiencyLevel.

AppointmentService

AppointmentService is the line-item entity that replaced a bare many-to-many relationship between Appointment and Service. Each row represents one service selected within one appointment, whether it is the core service or an add-on, which is what allows a single appointment to include several services (and their combined duration) rather than just one.

Its self-referencing relationship (“add-on for”) lets an add-on line-item point to the specific core-service line item it modifies, so the system can tell which core service each add-on belongs to within a multi-service appointment.

Summary

Our **business** operates at one **location**, and this location employs **staff**. Each staff member has **availability**, which generates **time slots**. The business organizes its offerings into **service categories** containing **services**, and staff are linked to the services they can perform through a **proficiency level**. A **customer** books an **appointment** for a service with a qualified staff member in an available time slot. The appointment is secured by a **payment**, made with one of the customer's stored **payment methods**, and triggers a **notification**. If the customer cancels, a **cancellation** record is created, and the business's **cancellation policy** determines any fee or refund.

2. Key Design Decisions

- **Separating SERVICE_CATEGORY and SERVICE.** Customers think in categories first (Nails, Pet Grooming) and then choose a specific service (Manicure). Splitting them into two entities supports this browsing flow, and it lets the business add or retire individual services without restructuring the catalog. Both are linked to the business so that each business controls its own offerings.
- **Using ProficiencyLevel as a many-to-many relationship between STAFF and SERVICE.** A staff member can perform many services, and a service can be performed by many staff, so a direct one-to-many link would not work. A many-to-many relationship also lets us store the proficiency level, which only makes sense for a specific staff-and-service pair. This ensures customers are matched with qualified staff.
- **Separating STAFF_AVAILABILITY from TIME_SLOT.** Availability is a recurring pattern (for example, Mondays 9am to 5pm), while a time slot is a concrete bookable instance. Keeping them separate means staff schedules are entered once and time slots are generated from them. Because each appointment reserves a specific time slot, the system can prevent double-booking.
- **Making APPOINTMENT the central entity.** Appointment is where the main parts of the system meet: customer, service, staff, time slot, payment, policy, and cancellation. Placing these connections on a single entity makes the booking transaction easy to trace and query.
- **Storing CANCELLATION separately from APPOINTMENT.** Appointment already has a Status attribute, but a status alone cannot record who cancelled, when, why, and what the outcome was

(for example, fee charged or refund issued). Since only some appointments are cancelled, a separate entity also avoids many empty fields on the appointment table and preserves a history for future reporting.

- **Tying CANCELLATION_POLICY to the business and linking it to appointments.** Each business has its own rules for notice periods, fees, and refunds, so the policy belongs to the business. Linking each appointment to the policy that governs it means the correct rule is applied even if a business changes its policy later.
- **Separating PAYMENT_METHOD and PAYMENT.** A payment method (a card on file) is stored once and reused, while a payment is an individual transaction. Separating them supports repeat customers and lets one appointment have several payments, such as a deposit, a final charge, or a cancellation fee. Only the card type, last four digits, and expiration date are stored, which avoids keeping sensitive card data in our database.
- **Including NOTIFICATION as an entity.** Recording the type, send time, and delivery method of each notification gives the business an audit trail of what customers were told and when. It also supports the confirmations and reminders that reduce missed appointments.
- **Linking CUSTOMER to their own data.** Customers are connected directly to appointments, stored payment methods, and notifications so that each person's booking history and preferences are available, supporting a faster experience for repeat customers.